

Table 1: Paired masked-vs-unmasked RQ₁ comparisons ($n=105$ CVEs). Δ is masked minus unmasked. Win/Tie/Loss = masked higher / equal / lower. p values are Holm-corrected Wilcoxon signed-rank tests within each metric; r is the matched-pairs rank-biserial correlation.

Model	Metric	Δ	p (Holm)	r	Win/Tie/Loss
Claude Sonnet 4.5	NOR	0.03	0.75	0.18	28/53/24
	LCNR	0.00	0.79	0.04	35/39/31
GPT-5.2	NOR	-0.06	0.94	0.01	24/53/28
	LCNR	-0.01	0.64	-0.19	27/49/29
DeepSeek-V3.2	NOR	+0.02	0.73	0.21	29/54/22
	LCNR	0.00	0.21	0.32	31/55/19
GPT-4o	NOR	-0.04	0.84	-0.13	28/53/24
	LCNR	-0.02	0.73	-0.15	25/54/26
Llama-3.3-70B	NOR	+0.16	0.01	0.39	53/27/25
	LCNR	+0.17	0.02	0.40	45/35/25

Identifier Masking Contamination Check

Replication Package Document for
Beyond Labels: Evaluating LLMs on Vulnerable-Path Reconstruction

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This document accompanies the replication package for the paper above. It reports the semantics-preserving identifier-masking experiment used to probe whether RQ₁ path-reconstruction scores reflect training-data memorization rather than reasoning over the provided code.

1 Identifier Masking Contamination Check

Because all evaluated CVEs are public, a model might partially recall a vulnerability from training data rather than reason over the provided code. To probe this risk, we re-run the RQ₁ oracle setting on a semantics-preserving masked version of CVEPATH.

Masking. We rename user-defined identifiers to opaque tokens (`Cls_####` / `Fn_####` / `v_####`), rename files to `file_####.ext`, and redact comments, Python docstrings, CVE IDs, and project-name tokens, while keeping language keywords and a blacklist of standard-library and framework source/sink APIs intact. Decoding and aggregation match Table I of the main manuscript.

Results. Table 1 reports paired masked-minus-unmasked differences. If Table I of the main manuscript were driven by memorized public identifiers or filenames, stripping those cues should degrade NOR and LCNR. It does not: for the four strongest models, Holm-corrected NOR p -values are all at least 0.73 and effect sizes are negligible to small. The only significant change is Llama-3.3-70B, which *improves* under masking (NOR +0.16, Holm $p=0.01$, $r=0.39$), the opposite of a memorization signature. The model ordering is preserved. We therefore find no evidence that the Table I numbers reflect identifier memorization on this setting.